

Fundamentals of Inference

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Overview

- Design-based framework
 - Some call “randomization inference”
- Inference under this approach
 - Depends on the characteristics of the sample design
- Questions about the framework

Review of Design-Based Framework

- Finite Population, Estimand, Design, and Estimation Procedures

Finite Population: $Y = \{Y_1, Y_2, \dots, Y_N\}$

Estimand: $Q(Y_1, Y_2, \dots, Y_N)$

Design: $\Pr(I_1, I_2, \dots, I_N)$ is the joint distribution of random binary variables $(I_1, I_2, \dots, I_N)$

The sample is a specific realization of $(I_1, I_2, \dots, I_N)$:

i.e., the set n I_j 's = 1 and the rest of the I_j 's = 0 corresponds to the sample of observed $y = \{y_1, y_2, \dots, y_n\} \subset Y$.

Estimating Procedure (estimator): $q(y)$

Sampling distribution of $q(y)$ is the distribution of the values of $q(y)$ across all possible samples.

$E(q(y))$ is the mean and $V(q(y))$ the variance of the sampling distribution $q(y)$.

Desirable $q(y)$ is unbiased or consistent for $Q(y)$ with small variance.

- For a particular sample, apply the estimation procedure:

Estimator of $Q(Y)$ is $q(y)$

Estimator $V(q(y))$ is $\hat{V}$

- If the procedure is “desirable”, then what is produced is “desirable”.

Large sample properties of the sampling distribution

- Assume that N is **large** and n is **large** but n/N is **small**
 - The design-based approach relies on **asymptotics** (large sample, large population), from Module 7

- For many estimators $q(y)$, the sampling distribution can be approximated as a standard normal distribution

$$\frac{q(y) - E(q(y))}{\sqrt{V(q(y))}} \sim N(0,1)$$

- A histogram of all possible estimates (collection of $q(y)$ across all possible repeated samples $y = \{y_1, y_2, \dots, y_n\}$) can be approximated by the density function of a standard normal distribution $N(0,1)$ (mean 0, variance 1)
- Above $\sim N(0,1)$ comes from a Central Limit Theorem for sampling from large finite populations with **large** samples.

Improve Normal Approximation (Delta method)

- Sometimes the normal approximation can be improved by using a transformation of $q(y)$.

In general, consider a transformation $g(\cdot)$ of $q(y)$, i. e., $g(q(y))$. Need to obtain $Var [g(q(y))]$.

- Taylor expansion of $g(q(y))$ about $q(y) = E[q(y)]$ to derive $Var [g(q(y))]$ and its estimator:

$$g(q(y)) = g(E[q(y)]) + \left[\frac{dg(q)}{dq} \right] \Big|_{q(y)=E[q(y)]} \times (q(y) - E[q(y)]) + \text{error}$$

$$g(q(y)) \approx g(E[q(y)]) + \left[\frac{dg(q)}{dq} \right] \Big|_{q(y)=E[q(y)]} \times (q(y) - E[q(y)])$$

$$[g(q(y)) - g(E[q(y)])]^2 \approx \left(\left[\frac{dg(q)}{dq} \right] \Big|_{q(y)=E[q(y)]} \right)^2 [(q(y) - E[q(y)])]^2$$

$$[g(q(y)) - E(g[q(y)])]^2 \approx \left(\left[\frac{dg(q)}{dq} \right] \Big|_{q(y)=E[q(y)]} \right)^2 [(q(y) - E[q(y)])]^2, \text{ using } E(g[q(y)]) \approx g(E[q(y)])$$

$$E[g(q(y)) - E(g[q(y)])]^2 \approx \left(\left[\frac{dg(q)}{dq} \right] \Big|_{q(y)=E[q(y)]} \right)^2 E[(q(y) - E[q(y)])]^2$$

$$\text{Thus, the } Var [g(q(y))] \approx \left(\left[\frac{dg(q)}{dq} \right] \Big|_{q(y)=E[q(y)]} \right)^2 Var [q(y)],$$

Improved Normal Approximation (Delta method) (cont.)

$$\text{Var} [g(q(y))] \approx \left(\left[\frac{dg(q(y))}{d(q(y))} \right] \Big|_{q(y)=E[q(y)]} \right)^2 \times V(q(y)), \text{ where } V(q(y)) = \text{Var}(q(y))$$

$$\frac{g(q(y)) - g(E[q(y)])}{\left(\left[\frac{dg(q(y))}{d(q(y))} \right] \Big|_{q(y)=E[q(y)]} \right) \times \sqrt{V(q(y))}} \sim N(0,1)$$

Example: Consider **natural log** transformation:

Let $g(q(y)) = \log(q(y))$

$$\text{Since } \left[\frac{d \log(q(y))}{d(q(y))} \right] \Big|_{q(y)=E[q(y)]} = \left[\frac{1}{q(y)} \right] \Big|_{q(y)=E[q(y)]} = \frac{1}{E[q(y)]}$$

$$\text{Then } \frac{E(q) [\log(q(y)) - \log(E(q(y)))]}{\sqrt{V(q(y))}} \sim N(0,1)$$

Interval Estimation

- So far we have focused on the procedures for **point estimation** (to produce a single estimate) of the unknown population quantity, $Q(Y)$
- Most times it might be useful to provide a **range for plausible values of $Q(Y)$** , rather than a single estimate.
- Procedures for creating such intervals: Confidence intervals

Sample: $y = \{y_1, y_2, \dots, y_n\}$

procedure: $(q_L(y), q_U(y)); \Pr(q_L(y) \leq Q \leq q_U(y)) = 1 - \alpha$

Apply to a particular sample: (q_L, q_U) is $100(1 - \alpha)\%$ confidence interval

- The procedure guarantees that $100(1-\alpha)\%$ of all possible intervals (across all possible samples) will include $Q(Y)$
- The most efficient interval construction procedure produce the shortest possible intervals

Large sample intervals

- Use the aforementioned **large sample** (standard normal) approximation of the sampling distribution
- Consider an unbiased estimation procedure:

$$E(q(y)) = Q(Y)$$

$$\frac{q(y) - Q(Y)}{\sqrt{V(q(y))}} \sim N(0,1)$$

z is such that

$$\Pr\left(-z \leq \frac{q(y) - Q(Y)}{\sqrt{V(q(y))}} \leq z\right) = 1 - \alpha$$

E.g.,

$$1 - \alpha = 0.95, z = 1.96$$

$$1 - \alpha = 0.90, z = 1.64$$

$$-z \leq \frac{q(y) - Q(Y)}{\sqrt{V(q(y))}} \leq z$$

$$\Rightarrow -z\sqrt{V(q(y))} \leq q(y) - Q(Y) \leq z\sqrt{V(q(y))}$$

$$\Rightarrow q(y) - z\sqrt{V(q(y))} \leq Q(Y) \leq q(y) + z\sqrt{V(q(y))}$$

$$\approx (q_L = q(y) - z\sqrt{\hat{V}}, q_U = q(y) + z\sqrt{\hat{V}})$$

Interpretation: For $z = 1.96$, 95% of the intervals from all possible samples will include the estimand of interest, $Q(Y)$.

For a particular sample

- Assume $z = 1.96$
- Get upper and lower 95% confidence limits: (q_L, q_U)
- This is **only one** of the possible confidence intervals as they change with differing samples. It could be one of the approximately 95% of the intervals that includes the estimand $Q(Y)$ or it could be one of the approximately 5% intervals that excludes the estimand $Q(Y)$
- Choose z so that you have sufficient confidence that (q_L, q_U) is a range of plausible values of the estimand $Q(Y)$.

Example

- A finite population of a geographical area of a city consists of $N=270$ blocks and a simple random sample (wor) of $n=20$ blocks is selected.
 - Goal: Estimate mean no. of housing units **renting** in each block for the finite population $Q(Y) = \bar{Y}$
 - Data for sampled blocks: 3, 5, 5, 52, 21, 34, 3, 0, 0, 0, 17, 14, 0, 0, 2, 54, 11, 11, 0, 23.
 - Sample mean, $q(y) = \bar{y} = 255/20 = 12.75$ rentals per block

- Unbiased estimate of the finite population variance, $S^2 = \sum_{i=1}^N (Y_i - \bar{Y})^2 / (N - 1)$ is:

$$s^2 = \sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1) = 278.62$$

- Estimated sampling variance of $\bar{y}$ is

$$\hat{V} = (1-20/270) \times s^2/20 = 12.90$$

- Standard error of $\bar{y}$ is $\sqrt{\hat{V}} = \sqrt{12.90} = 3.59$
- 95% confidence interval: $(12.75 - 1.96 \times 3.59, 12.75 + 1.96 \times 3.59) = (5.71, 19.79)$
- **Inference:** For $\alpha = 0.05$ (a 95% confidence interval), you would “reject” any hypotheses suggesting that the mean is outside of the confidence interval (5.71, 19.79)

Inference about population proportions

- Suppose that the survey variable, Y , is a binary or “dummy” variable.
- Example (from previous slide): **Whether or not a block has any rental dwelling units**
- **Estimand:** Population proportion of blocks with **no** rental units
 $Q(Y) = \sum_{j=1}^N Y_j / N$, where $Y_j = 1$ if block $j=1, 2, \dots, N$ has no rental units, and 0 otherwise
- **Design:** Simple random sample (wor) of size n

Estimating Procedure:

$$q(y) = \sum_{i=1}^n y_i / n$$

Variance estimator of $q(y)$

$$\hat{V} = (1 - f)q(y)(1 - q(y))/(n - 1),$$

$$f = n/N$$

95% Confidence interval:

$$(a - 1.96 \times \sqrt{\hat{V}}, a + 1.96 \times \sqrt{\hat{V}})$$

Computing 95% Confidence Interval:

$$q(y) = 6/20 = 0.3$$

$$\hat{V} = (1 - 20/270) \times (0.3 \times 0.7)/19 = 0.0102$$

$$SE = \sqrt{\hat{V}} = 0.1$$

95% Confidence Interval:

$$0.3 \pm 1.96 \times 0.1 = (0.1, 0.5)$$

Alternative approaches for constructing confidence intervals for proportions

- Transformation can improve the normal approximation
Two popular transformations for proportions:

$Q(y) = \text{Population proportion}$

$q(y) = \text{sample proportion}$

$V = \text{sampling variance of } q(y)$

$n = \text{sample size}$

$N = \text{population size}$

$f = n/N$

Arcsine-square root
transformation

$$\sin^{-1} \sqrt{q(y)} \sim N(\sin^{-1} \sqrt{Q(y)}, (1 - f)/(4n))$$

Logit transformation

$$\log \frac{q(y)}{1 - q(y)} \sim N\left(\log \frac{Q(y)}{1 - Q(y)}, \frac{V}{[Q(y)(1 - Q(y))]^2}\right)$$

- Compute the confidence interval on the transformed scale, and then re-transform to the original scale
- See Franco et al. (2019) on Canvas for other methods

Example

- Proportion of blocks with no rental units

$$q(y) = 6/20 = 0.3$$

$$\sin^{-1} \sqrt{0.3} = 0.5796 \text{ radians}$$

$$\text{Estimated sampling variance } (1 - f)/(4n) = 0.9259/80 = 0.0116$$

$$\begin{aligned} & 95\% \text{ confidence interval for } \sin^{-1} \sqrt{Q(Y)} \\ & (0.5796 - 1.96 \times \sqrt{0.0116}, 0.5796 + 1.96 \times \sqrt{0.0116}) \\ & = (0.3685, 0.7907) \end{aligned}$$

$$\begin{aligned} & 95\% \text{ confidence interval for } Q(Y) \\ & = ([\sin(0.3685)]^2, [\sin(0.7907)]^2) \\ & = (0.13, 0.51) \end{aligned}$$

Homework Exercise:

Compute a 95% confidence interval for $Q(y)$ using the log-odds (logit) transformation.

Statistical Significance

- **Restate the problem as a test of a hypothesis:** Municipality wants to know whether the average number of rentals per block > 10 ?
 - This is the cutoff for a government subsidy of rent
- Thus, the claim or hypothesis is that the average rental unit per block is > 10 .
 - *How do we test this claim statistically?*
- Two approaches
 - Construct confidence interval with a specific level of confidence and check whether the hypothesized value is contained in the interval
 - In the example, 95% confidence interval (slide 10) is (5.71, 19.79) which includes 10.
 - Set it up as significance test of the **null hypothesis**: $H_0: \bar{Y} \leq 10$ vs. **alternative hypothesis** $H_A: \bar{Y} > 10$

Significance Testing

- Example

Testing $H_0: \bar{Y} \leq 10$ vs. $H_A: \bar{Y} > 10$

For the boundary point, $\bar{Y} = 10$:

$$Z = \frac{q(y) - 10}{\sqrt{V(q(y))}} \approx \frac{q(y) - 10}{\sqrt{\hat{V}}} \sim N(0,1)$$

For the particular sample

$$\frac{12.75 - 10}{\sqrt{12.90}} = 0.77$$

Interpretation:

(1) Values of Z that are higher than 0.77 is favorable to $H_0: \bar{Y} \leq 10$.

(2) $\Pr(Z > 0.77 | H_0) = 0.24$ (p-value)

(3) Since 0.24 is not that “small”, i.e., < 0.05 (cannot reject H_0).

Sample Size Determination for Proportions

- How many units should be sampled?
- One approach is to specify the acceptable sampling variance
- Consider inference about proportions:

$$V = \frac{(1-f)Q(Y)(1-Q(Y))}{n-1}$$

$$V_{\max} = \text{Maximum}(V) = \frac{(1-f) \times \frac{1}{4}}{n-1} \text{ (why the } \frac{1}{4} \text{?)}$$

Goal is to choose n so that $V_{\max} \leq c$

Sample Size Determination for Means

- Simple random sample, sampling variance:

$$V = (1 - f)\sigma^2/n$$

Let $V \leq c$ then

$$n \geq \frac{1}{\frac{1}{N} + \frac{c}{\sigma^2}} \approx \frac{\sigma^2}{c}, \sigma^2 \text{ is population of } Y$$

- Need an estimate of σ^2 from a sample of a pilot or previous study.

Design Effects

- Often, sample designs are more complex than SRS, involving stratification, weighting, and clustering.
 - **Stratification**, generally, decreases the sampling variance
 - **Clustering** increases the sampling variance because of correlation of sampling indicators
 - **Weighting** may increase or decrease the sampling variance, depending upon correlation between weights and survey variables y
- The effect of a complex sample design on the variance of an estimate is measured using the **design effect** concept
- **Effective Sample Size**: size of an SRS that would give you the same precision

n = sample size

V_A = Actual Sampling variance
(under the complex sample design)

V_{SRS} = SRS sampling variance (the same sample size)

$$deff = \frac{V_A}{V_{SRS}}$$

Effective sample size, $n^* = n/deff$

Design Effects (Contd.)

- Simple Approximations

- For Clustering:

$c =$ *number of clusters sampled*

$k =$ *Average number of units sampled per cluster*

$n = ck,$ *number of observations.*

$\rho =$ *Intraclass correlation coefficient*

(Correlation of variable y between pairs of observations in the same cluster)

$$\frac{V_{cluster}}{V_{SRS}} = deff_{cluster} = 1 + (k - 1)\rho$$

(Why $k - 1$?)

Design Effects (Contd.)

- Simple *def* approximation for sample weighting (a.k.a. “Unequal Weighting Effect” or UWE)

$$\bar{w} = \text{mean of weights}, \frac{1}{n} \sum_{i=1}^n w_i$$

$$s_w = \text{standard deviation of weights}, \sqrt{\frac{1}{n-1} \sum_{i=1}^n (w_i - \bar{w})^2}$$

$$c = s_w / \bar{w}, \text{ the coefficient of variation of the weights}$$

$$def_{weight} = 1 + c^2$$

- **NOTE:** UWE assumes stratified sampling, independence of weights and key variables, and is an **upper bound**
- For planning purposes:
 - Develop the sample size for a simple random sample design
 - Inflate it by the design effects (see the Effective Sample Size, n^* , definition):

$$n^* = def_{weight} \times def_{cluster} \times n_{SRS}$$

Questions to Ponder that relate Design-Based Inference

- What if an analyst is working with data from a complex sample, and **ignores** the weighting, stratification, and cluster sampling in the estimation procedures?
 - Will 95% confidence intervals have the level of coverage of the true population parameter that they claim?
- What role does **sample stratification** play in sample size calculations? These simple design effect approximations only speak for weighting and cluster sampling.
- Do all intervals need to be 95% confidence intervals when using the design-based approach?
- What if we don't have large samples (or populations!) and we can't rely on the central limit theorem?

Questions to Ponder that relate Bayesian Modeling

- Design-based approach uses the properties of estimation and inference procedures across all possible samples to select a procedure and then applies that procedure to a sample at hand.
 - If the procedure is “good” then does it imply the result from applying that procedure to one particular sample is also “good”?
- I only have one sample. Why is it important or even relevant to consider all possible samples that I could have observed, but did not, when drawing inferences based on the sample in hand?
- There are two parts of the finite population: The sample I observe is what I know and the rest of the finite population is what I don't know.
 - Shouldn't the inference be conditional probability statements like, $Pr(\textit{What I don't know} \mid \textit{What I know})$?